lowThrust — how it works

A Kerbal Space Program mod (C#) that flies a low-thrust vessel to an orbit the player types in — about the same body, a moon, another planet or a planet's moon — within a delta-v / time budget, using the atmosphere where there is one. `lowThrust/mod/`.

1. The parts

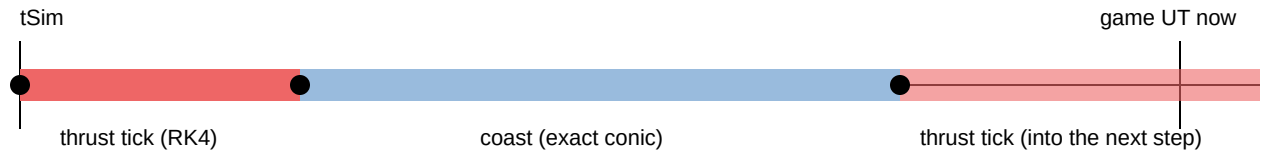
part	what it does
`core/` (no KSP)	The solver. Two-body maths and Lambert (`Orbit`), the within-SOI law (`Guidance`), the stand-in (`Sim`: a point mass, constant thrust, RK4), targeting and capture (`Transfer`), the atmosphere and the drag pass (`Aero`), the phase machine (`Mission`), the dv/time trade (`Tradeoff`), every tunable (`Config`).
`plugin/`	In the game. `Controller` flies the mission, on rails, and hands each pass through the air to the game's physics. `Game` reads the solar system (including each atmosphere's density) and the vessel through KSP's orbital elements, runs the engines on rails, estimates the drag area and sets the warp. `Attitude` turns and holds the vessel on rails with its reaction wheels, engine gimbal and RCS — only while a mission flies it (section 3a). `Window` is the toolbar window. `FlightLog` writes one log per flight.
`bridge/` (farm only)	A kRPC service, `LowThrustBridge`, that starts a flight and reports on it, for `tools/ltfly.py`. The mod itself needs no kRPC.
`tools/`, `tests/`	`lowthrust.exe mission|fly` (the stand-in on the command line; `--cda` gives the vessel a drag area so it aerobrakes), and the solver's tests (mono).

2. A flight, from the window

The player picks the body and fills any of periapsis, apoapsis, inclination, LAN and argument of periapsis (blank: free; one apsis alone: circular). Optionally Max Δv and Max time, and the highest warp the autopilot may use.

PLAN flies the whole mission in the stand-in over the trade-off grid (section 5) on worker threads and lists up to 10 options along the frontier. **START**: a Δv limit below `IMPULSIVE_MARGIN` × the impulsive estimate is refused at once; otherwise the engines are lit, SAS goes off, the throttle is shut, the attitude model is reset and the vessel's actuators measured (section 3a), the warp limits are relaxed, the drag area is estimated (section 7), the same scan runs (or the clicked option is used), and the first leg's `Mission` is built on a worker thread. **TERMINATE** stops it; warp off, every relaxed limit restored. With no mission flying, the mod changes nothing on any vessel.

3. Every physics step on rails



● a decision (Mission.DoStep) from the state there. Then: orbit written if anything thrust; attitude target set; warp chosen.

Figure 1. One physics step covers whatever game time the warp gives it; commands are decided as they fall due.

Attitude first. `Attitude.Step` turns or holds the vessel toward the target set at the end of the previous step (section 3a); the thrust axis it leaves, as components against the orbit (radial, along-track, normal), is the axis every thrust tick in this step flies.

Read. If the game's reference body differs from the planner's, the planner takes the game's state, the mission is told (`SoiChange`), and the step ends. Otherwise the state at tSim is read from the vessel's orbit elements, the mass from the vessel (so propellant any actuator drew is counted).

Decide when no command is running: engines and mass refreshed; `Mission.DoStep` returns a throttle, direction and tick, a pass (the craft is at the top of the air, inbound: section 7), or the end of the leg. A coast is run ahead in prediction until the mission next wants thrust or a pass, an SOI is near, or `COAST_AHEAD` of a revolution has gone; the span becomes one coast command, the next decision held ready.

Thrust command. It fires only if the thrust axis is within `ALIGN_DEG` when it starts; if not, the vessel coasts only as long as it needs to turn there (a bang-bang turn at 0.8 of the weaker pitch/yaw authority, plus 5 s), and the mission decides again. A fired command thrusts along the axis as the attitude leaves it each step; RK4 substeps of tick / `THRUST_SUBSTEPS`; every substep draws the propellants (thrust cut to the scarcest), lowers the mass, books the steering cost and checks for an SOI crossing. The throttle flown is handed to the attitude model for the gimbal's share.

Coast command. The exact conic, stopping at an SOI boundary. **Write.** If anything thrust, the vessel's orbit is set through `Orbit.SetOrbit`. This is the only place the mod writes an orbit.

Searches spread over steps. WAIT's moon-push search (up to ~80 trajectory evaluations) runs one evaluation at a time while the step has half its frame budget left, and goes on in the next steps while the coast continues; the warp is held to 5% of the time left to the push point meanwhile. It evaluates exactly what one uninterrupted search would.

Frames, engines, pointing

One right-handed inertial frame per body (x-y the equator, z north); states only through KSP's orbital elements; world space only for pointing, as components against the orbit. Each log opens with a rails check and a frame check. Engines: the lit `ModuleEngines`, vacuum thrust × limiter, Isp at 0 atm, propellant ratios from the part config, drawn through the part's flow rules. Pointing: the target is a burn's direction, or during a coast the next burn's direction where the coast ends. Each thrust substep books the dv lost to the axis being off the command.

3a. Turning on rails

On rails the game does not integrate rotation; `Attitude` does, for the vessel a mission is flying and no other.

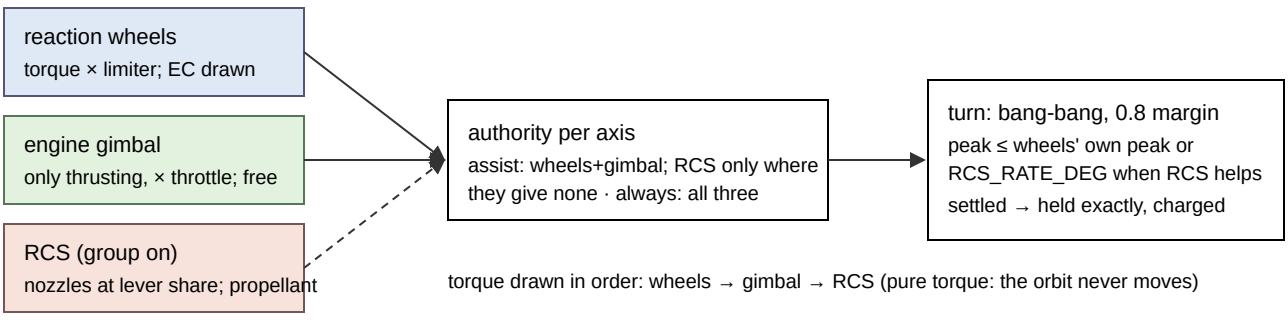


Figure 2. The actuators, what each costs, and how a turn uses them.

- **Measured off rails**, every step (read only, mission or not): the moment of inertia; each active reaction wheel's pitch/roll/yaw torque × authority limiter and its EC rate; each unlocked gimbal's torque at full thrust — every nozzle of its lit engine tilted through the range (× limiter, smallest of the per-direction ranges), axes masked by its pitch/roll/yaw toggles; each enabled RCS module's nozzles, the force

(`-up`, or `-forward`) about the centre of mass, axes masked. Per axis, RCS fires each nozzle at its lever's share of the longest lever (as the game does): the authority is the weaker direction's torque, and the thrust per torque costs the propellant (vacuum Isp, the module's mixture). Measured again at START, whatever the vessel's state.

- **Authority.** Free: wheels × the share of the needed EC available, plus gimbal × the throttle last flown (zero coasting or turning). RCS: only while the vessel's RCS action group is on and `RCS_MODE` is not off, × the share of its propellant available. `assist` (default): each axis turns on the free authority, on RCS only if the free one is zero (no wheels, no charge). `always`: free + RCS.
- **A turn**, integrated in 0.05 s substeps: the rate wanted is along the axis to the target, at most √(2 × 0.8 × authority × angle) and angle / 4 s, plus the target's own turn rate (measured between steps, measured afresh after every new command); the rate error closes over 0.5 s. When RCS adds authority the peak is also held to the larger of `RCS_RATE_DEG` and the wheels' own peak. Each substep's angular acceleration is taken from the wheels, then the gimbal, then RCS; EC and RCS propellant drawn for what each gave.
- **Held** once within 0.05° and the rate matched: the axis is put exactly on the target each step and charged — the torque that turns the rate with the target ($I\,d\omega/dt$) and holds it against the gyroscopic torque ($\omega \times I\omega$), wheels then gimbal then RCS; a hold they cannot give (torque, charge or propellant short) is let go and the vessel steered again.
- **Never the orbit.** Every actuator is pure torque. Each rotation snapshots the vessel's orbital elements and puts them back if they moved (counted as *orbit restores*). When the vessel comes off rails the rate turned is given back as a rigid rotation: every part spins at it and moves at $\omega \times r$ about the centre of mass, which sums to zero, so the momentum is the game's. With no mission, or a vessel not the mission's, nothing is turned, handed back or drawn.

4. The warp rate

The highest rails rate index such that one physics step (0.02 s × rate): keeps the autopilot within `FRAME_BUDGET_MS`; draws at most `STEP_RESOURCE_SHARE` of any propellant held; is at most 5 s while the vessel turns; is at most a quarter of the time left to the air (so the mission stops at its top), and 5% of the time left to a pending search's point; is allowed at the altitude (rails limits lowered to the atmosphere top or terrain + `TERRAIN_MARGIN_M`); stays under the game's own clamp before an SOI change (mirrored with margin, and a drop by the game held until the change has passed); is at most `MAX_WARP_INDEX` and the window's max warp. When the vessel is off rails (the game dropped the warp), it goes back on rails no faster than the air ahead allows — or, close above the air and coming down, the pass starts there. In a pass, physics warp is the highest of the game's physics rates at or below `AERO_PHYSICS_WARP` (a rate, ×4, so a mod that redefines the rate table does not change it).

5. Delta-v against time

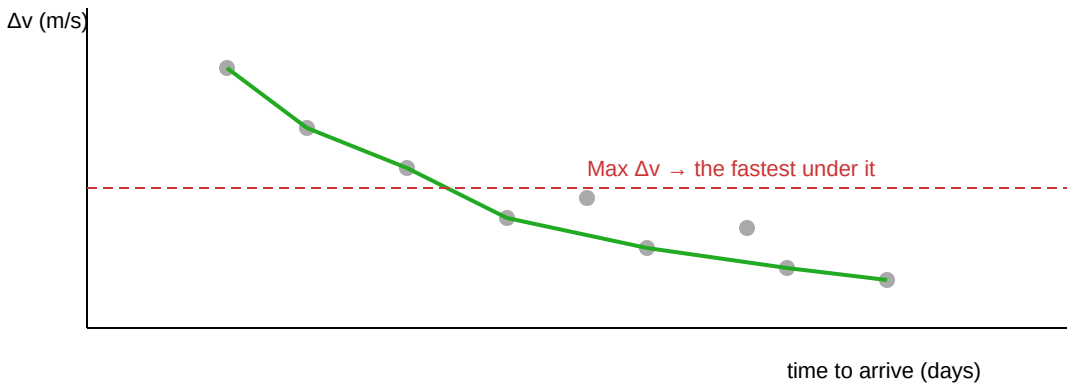


Figure 3. Each point is the whole mission flown in the stand-in at one arc width (and, to another planet, one transfer share). Green: the frontier.

Arc width: the guidance burns in arcs of half-width acos(η) round each apsis; `PLAN_GRID` (13) values of η. Narrow arcs are near-impulsive and cheap but take many revolutions; wide arcs are fast and dear. Transfer share (to another planet): the cheapest Lambert window arriving within `TRANSFER_TIME_SHARE` of the cheapest window's wait plus flight, at each of `TRANSFER_SHARES`. Max Δv → the fastest within it; max days → the cheapest within both; none → the fastest within `BALANCE_DV_SHARE` of the cheapest; a clicked option → that one; nothing → refused with what exists. The plan line's arcs is the chosen η. The stand-in plans with the vessel's drag area (aerobraking included, heating limit `AERO_HEAT_PLAN`).

6. The mission's phases

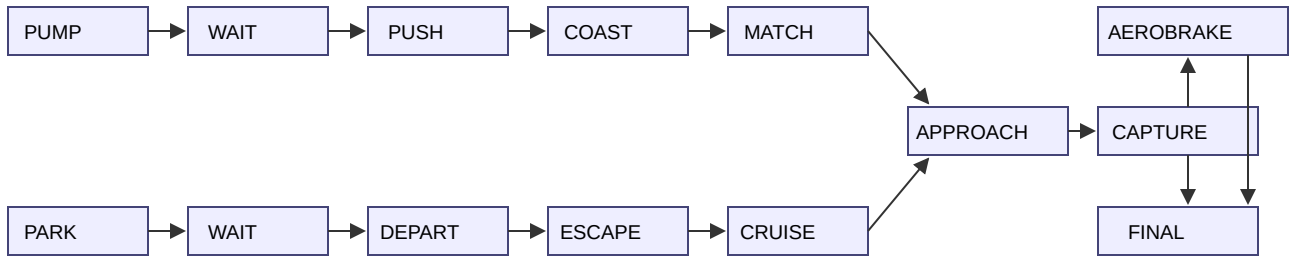


Figure 4. To a moon (top), to a planet (bottom); MATCH only when the arrival is too fast to capture. AEROBRAKE only where there is air; within one SOI: FINAL (→ AEROBRAKE → FINAL). Anywhere: TRIM (phasing to miss a moon, then back to the phase it left), the floor guard. Each is re-decided every tick from the current state.

phase	flies	ends when
PUMP / PARK / FINAL	`Guidance`: prograde/retrograde arcs on a held apse line, each fixing the opposite apsis (outward first); plane change at the nodes on the biggest orbit; arcs widened until the changes left fit `ARC_PASSES_MAX` passes, or `ETA_WIDE_MIN` while crossing another moon's band; each arc's burn capped so the next revolution meets no moon; a phasing trim once a revolution if one would. FINAL at a body with air and an apoapsis above the final one hands over to AEROBRAKE.	every targeted element within tolerance → WAIT, or done (FINAL)
WAIT (moon)	Coasts. At each push point (periapsis, or every 30° if no dearer than `PUSH_OFF_PERIAPSIS`) it solves the prograde push whose coast enters the moon's SOI on the aim (the finite burn; spread over steps, section 3); skips a pass that meets another moon or arrives too fast to capture, and remembers when the planned encounter is.	half the push before its point → PUSH
PUSH	Prograde, re-solved round its value each tick, looking at least 1.2 × as far ahead as the encounter WAIT planned.	< `CORRECT_DV` left → COAST
WAIT / DEPART (planet)	The window from the transfer share; waits for the periapsis pass nearest it (checked along the burn DEPART will fly), then burns prograde to the window's v∞.	v∞ reached or `DEPART_ARC_DEG` past periapsis → ESCAPE (or WAIT if still bound)
ESCAPE (planet)	Propagates the hyperbola to the SOI exit and takes the heliocentric state there. Once: if that state's closest approach to the target misses by less than the cruise would re-target (`RETARGET_MISS` SOI radii or `RETARGET_FRACTION` of its distance from the star, whichever is less), it coasts — the cruise's B-plane correction fixes it cheaper. Otherwise it aims at a Lambert arc (the cruise's re-target, solved from the exit) and burns inside the SOI, where the Oberth effect makes	SOI exit → CRUISE

	each m/s worth v∞/v of one at the exit: each tick the exit-velocity change the arc needs is mapped back to a burn now through the 3×3 sensitivity of the exit velocity to the current one (finite differences, 1 m/s per axis).	
COAST / CRUISE	B-plane correction toward the capture periapsis; Lambert re-target when the miss is too big to linearise (the same test as ESCAPE's).	SOI entry → APPROACH; MATCH due → MATCH
MATCH	ZEM/ZEV guidance about the parent onto a point outside the SOI, arriving at `ARRIVE_MARGIN` of the fastest capturable speed.	matched, `MATCH_END_S` before, or given up
APPROACH	On SOI entry at a body with air, the capture periapsis is moved into the air (section 7). Nudges the periapsis onto it; dodges a moon on the way in.	the capture law says burn → CAPTURE
CAPTURE	Burns centred on periapsis if it fits one pass; otherwise starts once the energy to shed reaches `CAPTURE_PATH_SHARE` of what full thrust removes on the path left, and throttles at that ratio. The energy the coming pass through the air sheds at its low end (drag ÷ margin) is left to it. Direction: retrograde above the middle of the periapsis band, its pull on the periapsis faded out smoothly toward the aim (5% above the floor at least); below the aim a lift along the periapsis gradient closes the error over the time left to periapsis (at most half the thrust; under the floor, ramped in over 5% of it, all of it). The direction never jumps between ticks.	bound below the capture apoapsis (or the moon-clear ceiling past periapsis) → AEROBRAKE where there is air, else FINAL
AEROBRAKE	Section 7	section 7's end rules → periapsis lifted out → FINAL
TRIM	A phasing burn so the next revolution misses a moon; in AEROBRAKE it is checked over the coast down to the next pass, before the pass.	done → the phase it left

A thrusting tick changes the velocity by at most `THRUST_DV_SHARE`; ticks are uniform in angle and short enough not to jump an SOI boundary — or the top of the air.

The floor guard (any phase): if the periapsis is below the safe floor, it is lifted. While aerobraking (AEROBRAKE, or a TRIM that returns to it) the floor is the deepest any pass may go (radius + the lowest safe pass altitude), and the lift goes to the planned pass depth if that is above it, else to 2 km above the air.

7. Aerobraking

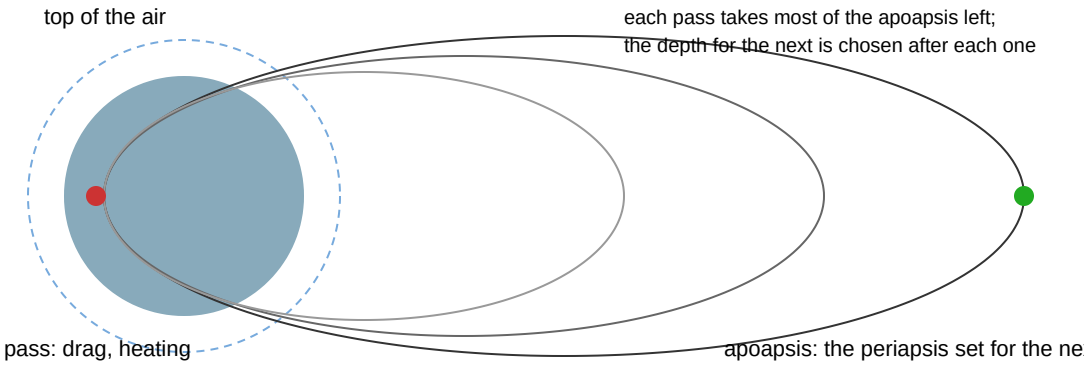


Figure 5. AEROBRAKE: passes through the air at periapsis, deep first and shallower as the apoapsis falls, the periapsis for the next one set by a burn round apoapsis.

The model (`Aero`)

- **Air**: density every 1 km from sea level to the top (0 there), log-linear between. In the game it is sampled from the body (`GetDensity(GetPressure, GetTemperature)` at the base temperature); the stand-in's tables were sampled from the game. The air turns with the body.
- **A pass** (`Aero.Pass`): RK4 from the top down and out — gravity plus drag q × cdA × k / m against the turning air, no thrust; step 500 m of path. It reports the energy shed, the drag Δv, the heating index H = $\sqrt{\rho}\, v^3 / 10^6$ at its peak and integrated (the load), the peak deceleration and the lowest altitude.
- **The deepest pass** (`Aero.Deepest`, bisection on the periapsis). Flown at `AERO_DRAG_MARGIN` × the drag and the density it must stay above the highest terrain + `AERO_TERRAIN_MARGIN_M` and `AERO_MIN_ALT_SHARE` of the air's depth; keep H and the load under the limits learned; decelerate at most `AERO_ACCEL_MAX`; and leave a periapsis the engine can lift to `AERO_LIFT_MARGIN_M` above the air within ±60° of the next apoapsis. Separately, flown at the apoapsis margin × the drag (density as modelled), it must leave an apoapsis at or above the final one (or stay unbound). The apoapsis margin is `AERO_RA_MARGIN` or the worst ratio seen between a pass's modelled and measured energy shed, whichever is larger, and at most `AERO_DRAG_MARGIN`^{2}. So the early passes are bound by heating and go as deep as it allows; as the apoapsis falls each pass is bound by the floor and takes most of what is left, shallower each time, the last a skim.

Capture

On SOI entry the capture periapsis becomes the deepest pass safe both for the arrival (its energy) and for the orbit the capture burn will leave (apoapsis the capture's). The capture law (section 6) leaves to the pass the energy it sheds at the low end.

AEROBRAKE

- After each pass (and at the start), the depth for the next: the deepest, or, if its next revolution meets a moon (checked at the modelled drag and at the margin above and below), the first shallower one in tenths of the way to the top that does not; if none does but the orbit as it is stays clear a revolution more, the periapsis is kept out of the air for one revolution.
- The guidance sets it: target apoapsis free, periapsis the depth, the target plane carried (so the plane change happens at the high apoapsis); the burn is the arc round apoapsis. The depth burn is not capped by the drag-free moon check the other phases use (the pass changes the orbit that check would predict). If the coast down to the pass meets a moon, a phasing TRIM is flown first.
- If a moon's SOI is entered and left again, the depth is re-planned on the orbit the flyby left; the floor guard (section 6) lifts a periapsis the flyby put below the deepest safe pass.
- Ends, whichever first: the burns since the last pass (depth burns, trims) cost more Δv than that pass shed at periapsis; or — once the heating limit has stopped growing — the next pass would lower the apoapsis by less than `AERO_MIN_GAIN` of what is left above the final one, or a burn at periapsis would finish the apoapsis for less than `AERO_MIN_PASS_DV`. Then the periapsis is lifted out of the air in the arc round apoapsis, widened (up to ±60°) to fit the lift, and FINAL shapes the orbit.

Learning, in the game

- The heating allowed starts at `AERO_HEAT_FIRST`. After each pass, every part's rise of max(skin ÷ skin maximum, interior ÷ maximum) from its value at the pass's start is extrapolated linearly to `AERO_HEAT_TARGET` of the maximum; the smallest factor over the parts (at most `AERO_HEAT_GROWTH`, at least 0.2) times the pass's modelled H and load are the limits for the next.
- The drag scale k = the energy the pass shed ÷ the model's at k = 1 from the same entry state; the ratio of measured to predicted energy shed (either way up) is kept as the observed miss for the apoapsis margin. The drag area starts from the parts' drag cubes (Mach 6, flow on the held nose).
- A pass not flown in the air (the vessel came off rails already past it) teaches nothing. The stand-in plans at `AERO_HEAT_PLAN` and learns nothing.

Flying a pass

The mission stops at the top of the air, inbound (the warp's steps shrink to it). The controller drops the warp; once the game has the vessel off rails it sets physics warp (section 4) and SAS to `AERO_HOLD`, and measures every step: H, the load, the dynamic pressure, the drag (the parts' `dragScalar`) and so the drag area, the deceleration, every part's temperatures. Out of the air, climbing, the temperatures are watched until none has risen for 20 s (at least 10 s, at most `AERO_SOAK_S`); then SAS goes off, the orbit is read back, the mission is told what the pass shed and how hot it ran, and the vessel goes back on rails. Nothing thrusts in a pass; off rails the attitude model only measures.

8. The log

`GameData/LowThrust/PluginData/logs/flight-<date>.log`: the `config:` line and defaults fingerprint, the rails check, the target, the vessel (mass, thrust, Isp, the wheels' angular acceleration per axis), `attitude on rails:` (gimbals and their authority at full thrust, RCS nozzles and authority, `RCS_MODE`, whether the RCS group is on, the RCS lever and Isp, the moment of inertia), the frame check, the propellants, the drag area (where passes may be flown), the `plan:`; the mission's events (among them *escape aimed at a Lambert arc …*, each *pass N planned …* and *aerobraking done (why)*) and telemetry at most once per `LOG_INTERVAL_UT`, each line ending with the EC the attitude has drawn and the RCS propellant (total / while thrusting / holding); for each pass its plan (depth, apoapsis before and after, drag Δv, heating and limit, load, deceleration), its entry (periapsis, speed, how far the game was ahead and at what warp), a line every 30 game seconds in the air, and its result (lowest altitude, apoapsis change, energy shed against the model, drag scale, heating limits, measured drag area and the hottest part's rise); at the end the result, wall and game seconds by phase, the autopilot's time per step, thrust ticks fired and not aligned, propellant shortfalls, the wheels' EC and the RCS propellant (holding, while thrusting), holds the actuators could not give, orbit restores, and the `steering:` line.